

Problem 1: Find the Largest Element in an Array

Sample Input:

5
10 25 8 45 12

Sample Output:

Largest element is 45

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int largest = arr[0];

        for(int i = 1; i < n; i++) {
            if(arr[i] > largest) {
                largest = arr[i];
            }
        }

        System.out.println("Largest element is " + largest);
    }
}
```

Problem 2: Find the Smallest Element in an Array

Sample Input:

5
10 25 8 45 12

Sample Output:

Smallest element is 8

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
        }

        int smallest = arr[0];

        for(int i=1;i<n;i++) {
            if(arr[i] < smallest) {
                smallest = arr[i];
            }
        }

        System.out.println("Smallest element is " + smallest);
    }
}
```

Problem 3: Find the Sum of Array Elements

Sample Input:

5
10 20 30 40 50

Sample Output:

Sum = 150

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        int sum = 0;

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        System.out.println("Sum = " + sum);
    }
}
```

Problem 4: Find the Average of Array Elements

Sample Input:

5
10 20 30 40 50

Sample Output:

Average = 30.0

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        int sum = 0;

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        double avg = (double) sum / n;
        System.out.println("Average = " + avg);
    }
}
```

Problem 5: Count Even Numbers

Sample Input:

5
1 2 3 4 5

Sample Output:

Even Count = 2

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        int count = 0;

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
            if(arr[i] % 2 == 0) {
                count++;
            }
        }

        System.out.println("Even Count = " + count);
    }
}
```

Problem 6: Count Odd Numbers

Sample Input:

5
1 2 3 4 5

Sample Output:

Odd Count = 3

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        int count = 0;

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
            if(arr[i] % 2 != 0) {
                count++;
            }
        }

        System.out.println("Odd Count = " + count);
    }
}
```

Problem 7: Reverse an Array

Sample Input:

5
1 2 3 4 5

Sample Output:

5 4 3 2 1

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
        }

        for(int i=n-1;i>=0;i--) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

Problem 8: Search an Element in an Array

Sample Input:

5
10 20 30 40 50
30

Sample Output:

Element Found

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
        }

        int key = sc.nextInt();
        boolean found = false;

        for(int i=0;i<n;i++) {
            if(arr[i] == key) {
                found = true;
                break;
            }
        }

        if(found)
            System.out.println("Element Found");
        else
            System.out.println("Element Not Found");
    }
}
```


Problem 9: Find Maximum and Minimum Element

Sample Input:

5
10 25 8 45 12

Sample Output:

Maximum = 45
Minimum = 8

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
        }

        int max = arr[0];
        int min = arr[0];

        for(int i=1;i<n;i++) {
            if(arr[i] > max) max = arr[i];
            if(arr[i] < min) min = arr[i];
        }

        System.out.println("Maximum = " + max);
        System.out.println("Minimum = " + min);
    }
}
```

Problem 10: Find the Second Largest Element

Sample Input:

5
10 20 30 40 50

Sample Output:

Second Largest = 40

CODE:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i=0;i<n;i++) {
            arr[i] = sc.nextInt();
        }

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        for(int i=0;i<n;i++) {
            if(arr[i] > largest) {
                secondLargest = largest;
                largest = arr[i];
            } else if(arr[i] > secondLargest && arr[i] != largest) {
                secondLargest = arr[i];
            }
        }

        System.out.println("Second Largest = " + secondLargest);
    }
}
```